Pretraining on unlabeled speech makes low-resource ASR practical

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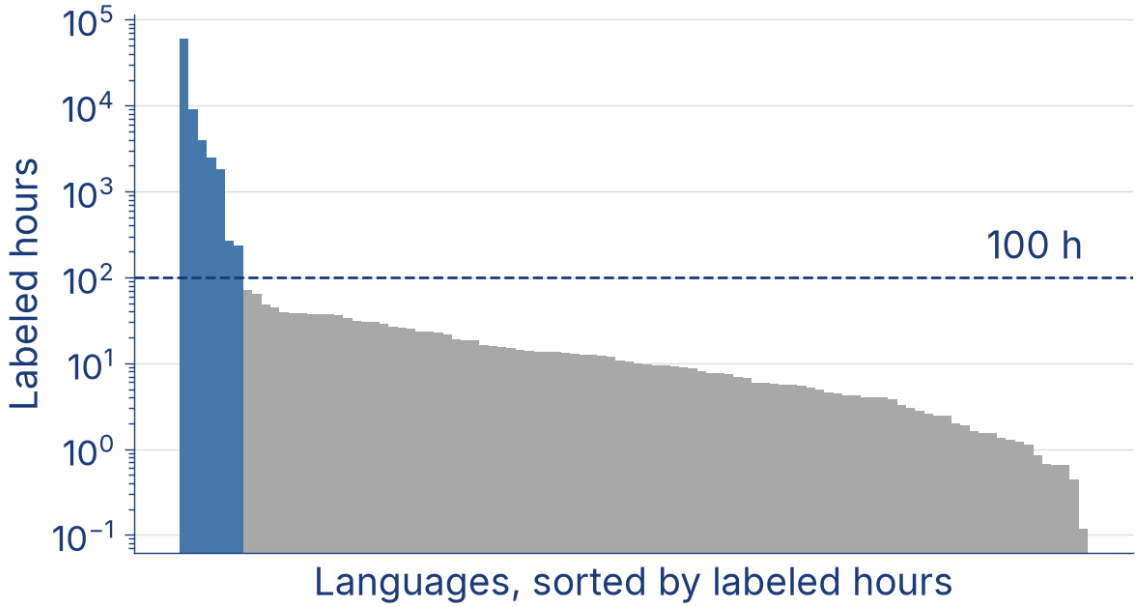
Method

Supervised ASR needs labels that most languages don't have

- **Labels are scarce:** 5 of 100 languages have over 100 h of transcripts[1]
  - Transcribing 1 h of speech takes about 8 h of expert time
- **Audio is not:** radio and podcasts exist in every language
- **Supervised ASR stalls:** over 40% WER with 10 h of labels[2]
  - Errors double again on accented and noisy speech

[1] Author et al. (2024), Venue. [2] Author et al. (2023), Venue.

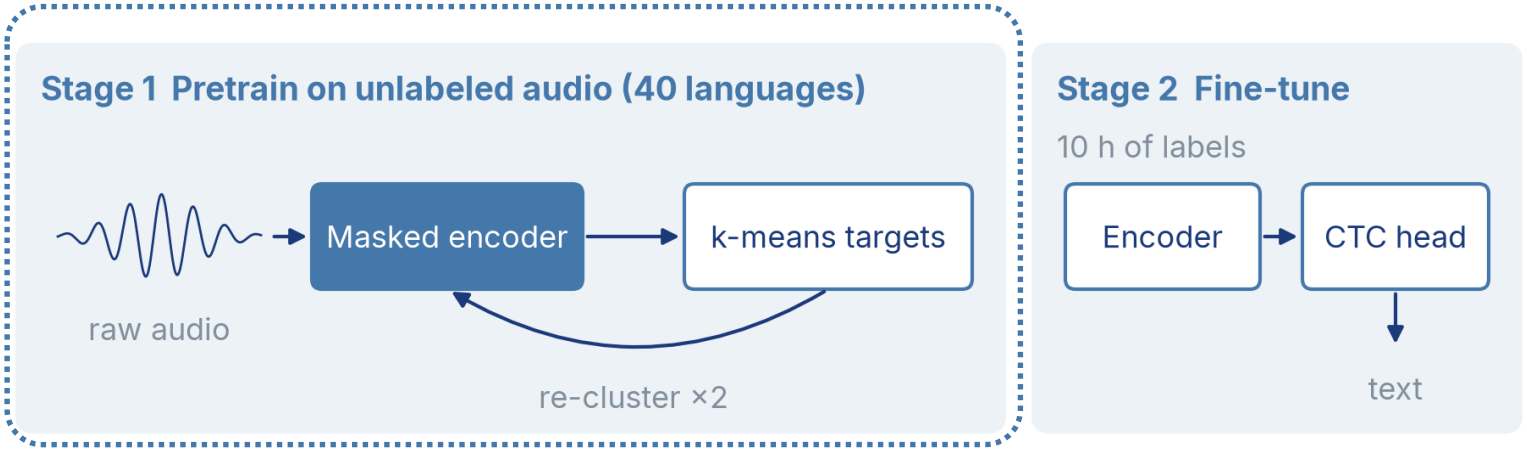
Only a handful of languages have enough labeled speech



- 5 of 100 languages have over 100 h of transcripts[1]
- Every language has unlabeled audio, from radio to podcasts

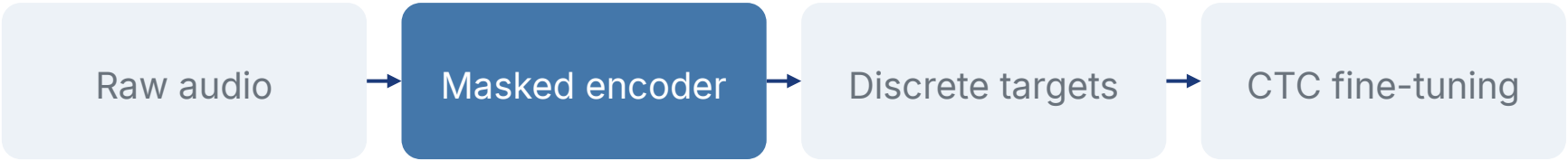
[1] Author et al. (2024), A census of speech corpora, Venue.

Stage 1 learns from audio alone



- **Mask:** spans of 10 frames, 8% of starts
- **Predict:** each masked frame's cluster
- **Re-cluster:** 100 MFCC clusters, then 500

Training runs in four steps, and only the last one uses transcripts



Steps 1 to 3 run once per language family; step 4 per language.

The loss predicts the targets of masked frames from their context

$$L = \sum_{t \in M} -\log p(z_t \mid x_{\setminus M}) + \lambda \|\theta\|^2$$

M masked frames (8% of starts, spans of 10)

z_t k-means cluster of frame t, from the previous iteration

λ weight decay; 0.01 in all runs

Re-clustering cannot make the targets worse

Proposition 1

If the encoder lowers the masked-prediction loss, re-clustering its features never raises it:

$$L(\theta_{k+1}; z^{(k+1)}) \leq L(\theta_k; z^{(k)})$$

- So iterations can only help
- Proof in Appendix B

Ours has the lowest WER on three of four test sets

	LibriSpeech			
Method	clean	other	CV	FLEURS
Supervised	8.4	19.2	21.7	24.1
Contrastive [1]	4.6	10.8	17.9	19.8
Masked prediction [2]	**3.9**	9.6	16.9	18.7
Multilingual fine-tuning [3]	5.3	12.4	16.1	17.5
Ours	4.0	**9.1**	**15.2**	**16.6**

WER (%), lower is better; best per column in dark navy. CV: Common Voice.

The paper makes three contributions of different kinds

Dataset

8K hours of unlabeled speech in 40 languages, all of it public

Objective

Masked prediction of discrete targets, no negative pairs

Analysis

When and why pretraining helps below 10 hours of labels

Code, data and models: example.com/project